

POWER, SPEED, AND STRENGTH: HOW THEY ACTUALLY RELATE

How power, speed, and strength build on each other.

Ask ten people what makes an athlete "explosive" and you'll get ten different answers, some say strength, some say speed, some say raw talent. The real answer is that power, speed, and strength are three distinct qualities that build on each other in a specific order, and understanding that order is what separates a training plan that actually makes you more athletic from a random collection of hard workouts.

Three Different Qualities, Often Confused

- **Strength** is your maximum ability to produce force, regardless of how fast, think a heavy, grinding deadlift near your limit. Slow, maximal, and foundational.
- **Power** is how quickly you can produce force, strength expressed fast. A vertical jump, a med-ball throw, an Olympic lift variation. Power is really "strength × speed" happening in a fraction of a second.
- **Speed** is how quickly you can move your body through space, sprinting, changing direction. Speed depends on power (you can't sprint fast without producing force quickly against the ground) but adds a large technical, coordination-based component on top of it.

They're related, but they are not the same trainable quality, and maxing out one doesn't automatically max out the others. A powerlifter with an enormous squat isn't automatically a fast sprinter, the strength is there, but the ability to express it **quickly** and **in the specific pattern sprinting requires** has to be trained separately.

Why Strength Comes First: But Isn't the Goal

Here's the relationship that actually matters for how a program gets built: **strength is the ceiling on power, and power is the ceiling on speed.**

You cannot express force quickly that you don't have in the first place, a certain baseline of strength is what gives power work something to work with. This is why every performance program, including Echelon's, includes real strength training (squats, presses, pulls, hinges) even though the end goal is power and speed, not a bigger max. Strength without power work stays slow. Power work without a strength base has a low ceiling. You need both, sequenced correctly.

But strength is a means, not the end point, in a performance program. Once a client has "enough" strength to support the power and speed qualities they need for their sport or goal, chasing more max strength for its own sake stops being the highest-value use of training time, that time is better spent on the power and

speed work strength was supposed to be feeding in the first place.

The Continuum, in Practice

Think of it as a chain: **Strength** → **Power** → **Speed**, each one built on the quality before it, each one requiring different training methods:

| Quality | Trained with | Rep speed | Load | |---|---|---|---| | Strength | Squats, deadlifts, presses, pulls | Controlled, near-maximal effort | Heavy (roughly 80%+ of max) | | Power | Jumps, throws, Olympic lift variations, trap-bar jumps | Maximal intended speed on every rep | Light-to-moderate, speed of movement matters more than the number on the bar | | Speed | Sprints, agility drills, change-of-direction work | Maximal effort, full recovery between reps | Bodyweight, or resisted lightly (sled, band) |

Notice that power and speed work is trained at *lower* loads than pure strength work, but with far higher intent and far more rest between reps, the goal isn't fatigue, it's repeated maximal-quality output.

What This Means for You

- Power and speed are built on top of a strength foundation, that's why strength work stays in a performance program even though it isn't the end goal.
- More max strength doesn't automatically make you faster or more explosive, it raises the ceiling, but power and speed still have to be trained directly.
- Power and speed work is trained fast and fresh, with real recovery between reps, not for fatigue or volume.
- A program that's "strength-only" or "conditioning-only" is missing a piece of the actual athletic-performance picture, real performance training deliberately sequences all three.

This article is educational and general in nature, not individualized coaching advice. Power and speed training (jumps, sprints, explosive lifting) carry real injury risk if attempted without an adequate strength base or sound technique, work with a qualified coach and talk to your doctor before beginning explosive training if you're new to it or returning from injury.